

Lab 2 Report: Version Control, VectorBoardObjects, and MVC Design

1. Introduction

This report documents the work completed for Lab 2 of the Intelligent Agents and Simulation coursework. The purpose of this lab is to extend the existing Pac-Man C++ project by adding new object-oriented components, using version control, documenting the code, and analysing the project structure through design diagrams.

The original project already contains several board object classes, including **BoardObject**, **Cookie**, **Player**, and **Wall**. In this lab, the project was extended by adding a new vector-based collection class called **VectorBoardObjects**, as well as two new game object classes: **Monster** and **Bomb**.

The main objectives of this lab are:

- To understand the role of version control in software development.
- To implement a vector-based alternative to ListBoardObjects.
- To limit the vector collection to a maximum of four objects.
- To add Monster and Bomb as new BoardObject subclasses.
- To test the new implementation using a separate test program.
- To document the code using Doxygen-style comments.
- To represent the project structure using UML and MVC diagrams.

2. Version Control

Version control is an important part of software development because it allows developers to track changes, manage different versions of a project, collaborate with others, and recover previous versions when necessary.

Although the practical material introduces Subversion, this lab was submitted through GitHub using a shared repository link. GitHub provides similar version-control benefits, including commit history, file tracking, remote access, and shared project visibility.

The GitHub repository was created for Lab 2 and Lab 3 submission. The Lab 2 files are organised inside a dedicated Lab 2 folder, while Lab 3 can be added later in a separate folder.

The repository structure is planned as follows:

IntelligentAgents_Lab2_Lab3/

```
|— Lab2/
|   |— PackMan.d/
|   |— screenshots/
|   |— diagrams/
|   |— README.md
|   |— Lab2_Report.pdf
|— Lab3/
```

Using GitHub makes the project easier to share with the assessor and also provides evidence of the uploaded files and project updates.

3. Source Code Extension

The main source-code work was completed in the **PackMan.d** folder of the Lab 2 project. The original files were preserved in a separate original folder, while the modified version was edited in the working folder.

The following files were added:

VectorBoardObjectsH.h
VectorBoardObjects.cpp
MonsterH.h
Monster.cpp
BombH.h
Bomb.cpp
TestVectorBoardObjects.cpp

The following existing files were modified:

BoardObjectH.h
Makefile.make

The purpose of these changes was to extend the object-oriented structure of the Pac-Man project without rewriting the entire system. The new classes follow the existing design style of the project and reuse the **BoardObject** base class.

4. VectorBoardObjects Implementation

The new class **VectorBoardObjects** was implemented as a vector-based collection of **BoardObject*** pointers. It provides a similar interface to the existing **ListBoardObjects** class, but it uses **std::vector** internally instead of **std::list**.

The class was declared in:

VectorBoardObjectsH.h

and implemented in:

VectorBoardObjects.cpp

The main private data members are:

```
std::vector<BoardObject*> list
std::vector<BoardObject*>::iterator internalIterator
MAX_OBJECTS = 4
```

The maximum number of stored objects is limited to four. This was implemented by checking the size of the vector before adding a new object.

The implemented methods include:

```
theFirst()
theNext()
addOne()
removeOnePlayer()
isThereAWall()
isCookieValue()
size()
isFull()
```

The method **addOne()** only adds a new object if the vector is not full. If the vector already contains four objects, the new object is rejected and a message is printed.

This design makes it possible to test whether the original list-based object collection can be replaced by a vector-based implementation with only localised changes.

5. Monster and Bomb Implementation

Two new board object classes were added:

Monster
Bomb

Both classes inherit from the existing **BoardObject** base class.

The Monster class was implemented in:

MonsterH.h
Monster.cpp

The Bomb class was implemented in:

BombH.h
Bomb.cpp

The BoardObjectH.h file was modified to add two new object types:

A_MONSTER
A_BOMB

The updated enum became:

```
enum ObjectTypes { THE_BACKGROUND, A_COOKIE, A_PLAYER, A_WALL,  
A_MONSTER, A_BOMB };
```

The **Monster** class represents an enemy object, while the **Bomb** class represents a special board object. In this lab, both classes have simple behaviour. Their **touch()** methods change their colour to demonstrate that they can behave as independent subclasses of **BoardObject**.

This implementation shows that the existing object-oriented design can be extended with new game objects without changing the whole project structure.

6. Compilation and Testing

A separate test program was created:

TestVectorBoardObjects.cpp

This test program verifies the functionality of **VectorBoardObjects**, **Monster**, and **Bomb**.

The **Makefile.make** file was updated to compile the test program. A new object list was added:

```
TEST_OBJS = TestVectorBoardObjects.o VectorBoardObjects.o Cookie.o  
Player.o Wall.o Monster.o Bomb.o
```

A new build target was also added:

```
TestVectorBoardObjects : $(TEST_OBJS)
g++ $(DBG) $(TEST_OBJS) -o TestVectorBoardObjects
```

The test program was compiled using:

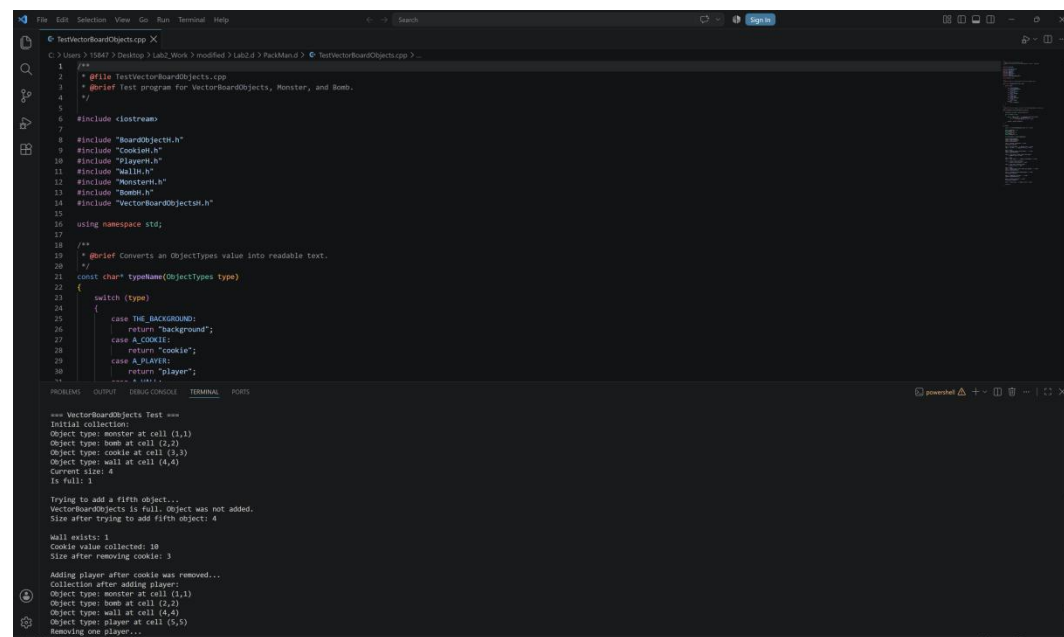
```
mingw32-make -f Makefile.make TestVectorBoardObjects
```

It was then executed using:

.\TestVectorBoardObjects.exe

The test output confirmed the following results:

- A monster object was added successfully.
- A bomb object was added successfully.
- A cookie object was added successfully.
- A wall object was added successfully.
- The collection reached size 4.
- The collection reported that it was full.
- A fifth object was rejected.
- A wall was detected.
- A cookie value was collected.
- A player was added after the cookie was removed.
- The player was removed successfully.
- The final collection contained the expected remaining objects.



```
1 //
2 * @brief TestVectorBoardObjects.cpp
3 * @brief Test program for VectorBoardObjects, Monster, and Bomb.
4 */
5
6 #include <iostream>
7
8 #include "BoardObject.h"
9 #include "Cookie.h"
10 #include "Player.h"
11 #include "Wall.h"
12 #include "Monster.h"
13 #include "Bomb.h"
14 #include "VectorBoardObjects.h"
15
16 using namespace std;
17
18 /**
19 * @brief Converts an ObjectTypes value into readable text.
20 */
21 const char* typeName(ObjectTypes type)
22 {
23     switch (type)
24     {
25     case THE_BACKGROUND:
26         return "background";
27     case A_COOKIE:
28         return "cookie";
29     case A_PLAYER:
30         return "player";
31     }
32 }
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7. UML Class Diagram

A UML class diagram was created to represent the structure of the extended Lab 2 project.

The diagram includes the following main classes:

BoardObject

Cookie

Player

Wall

Monster

Bomb

ListBoardObjects

VectorBoardObjects

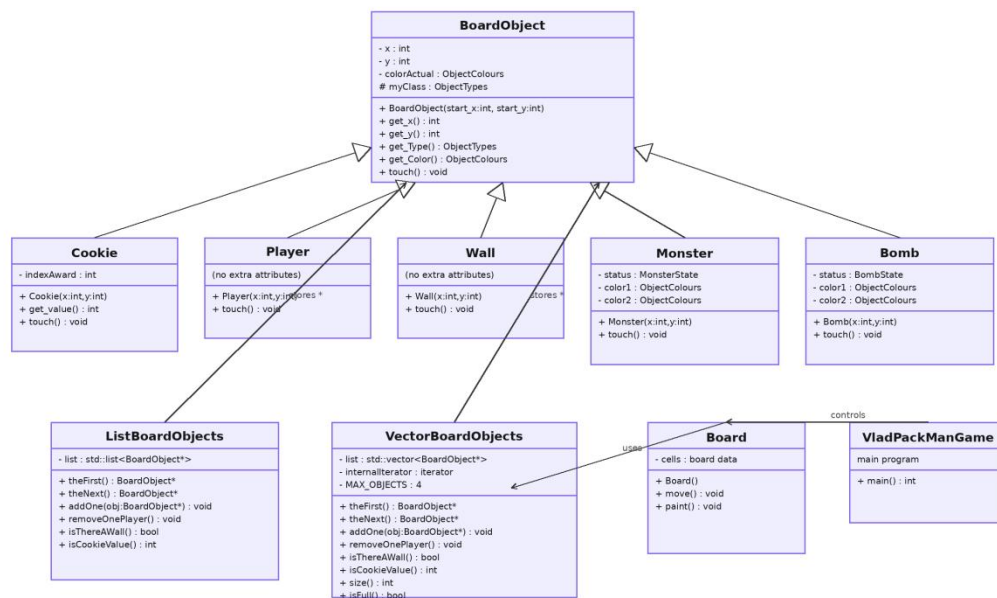
Board

VladPackManGame

The diagram shows that **Cookie**, **Player**, **Wall**, **Monster**, and **Bomb** all inherit from **BoardObject**. This inheritance structure allows all these objects to be treated through the same base class type.

The diagram also shows that both **ListBoardObjects** and **VectorBoardObjects** store pointers to **BoardObject** objects. This is important because it allows the collection to store different types of game objects through a common interface.

Lab 2 UML Class Diagram



Generalization arrows point to BoardObject. ListBoardObjects and VectorBoardObjects store BoardObject pointers.

8. MVC / Sequence Diagram

An MVC and sequence diagram was also created to explain the overall project structure.

The project can be interpreted using the Model-View-Controller pattern:

Model:

BoardObject, Cookie, Player, Wall, Monster, Bomb, Board, ListBoardObjects, VectorBoardObjects

View:

Terminal output and board painting behaviour

Controller:

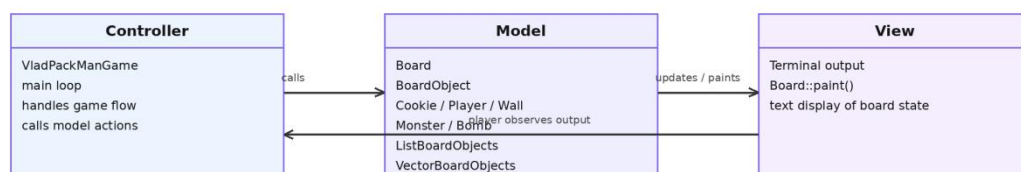
VladPackManGame and the main program flow

The model contains the game data and game objects. The view is responsible for displaying the game state. The controller coordinates the game execution and calls the appropriate model and view operations.

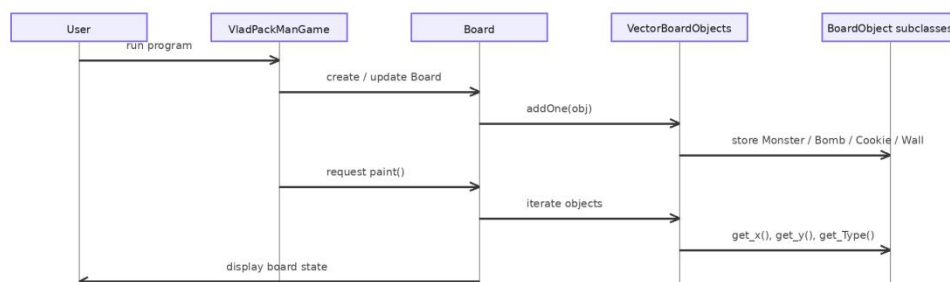
The sequence diagram shows a simplified execution flow:

1. The user runs the program.
2. VladPackManGame starts the game flow.
3. The Board manages the game state.
4. VectorBoardObjects stores BoardObject-derived objects.
5. The program retrieves object information.
6. The board state is displayed to the user.

Lab 2 MVC / Sequence Diagram



Simplified Execution Sequence



9. Doxygen Documentation

Doxygen-style comments were added to the new source files and header files. These comments describe the purpose of each class and method.

The following files include Doxygen-style comments:

VectorBoardObjectsH.h
VectorBoardObjects.cpp
MonsterH.h
Monster.cpp
BombH.h
Bomb.cpp
TestVectorBoardObjects.cpp

Examples of documented elements include:

- **Class descriptions**
- **Constructor descriptions**
- **Method descriptions**
- **Parameter descriptions**
- **Return value descriptions**

This documentation improves code readability and makes it easier for another developer to understand the purpose of the new classes. It also supports future automatic documentation generation using Doxygen.

10. Conclusion

Lab 2 extended the original Pac-Man C++ project by adding a new vector-based object collection and two new board object subclasses. The **VectorBoardObjects** class provides a similar interface to **ListBoardObjects**, but it stores objects using **std::vector** and limits the collection to four objects.

The new **Monster** and **Bomb** classes demonstrate how the project can be extended using inheritance. Both classes inherit from **BoardObject** and can be stored in the same object collection as existing objects such as **Cookie**, **Player**, and **Wall**.

The test program confirmed that the new vector collection works correctly. It successfully added objects, rejected a fifth object when full, detected a wall, collected

a cookie value, added a player, removed a player, and preserved the expected remaining objects.

The UML and MVC diagrams helped explain the structure of the project and the relationship between different classes. Overall, this lab improved understanding of C++ object-oriented design, collection classes, version control, build automation, and software architecture documentation.